POE-Part 2

ST10433945 – NEHA SINGH – GPR 1

ST10439309 – MOFENYI KUBHEKA – GPR 2

ST10436124 – OARABILE MARWANE – GPR 1

ST10448763 – LESEGO RAPHAHLA MAROTA – GPR 1

ST10443515 – MAMAPHALE LEAGO TEMA – GPR 2

PROG7313

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1. Project Links

All source code, build artefacts and continuous integration workflows associated with the Zentavo prototype are hosted on GitHub. A demonstration video, published as an unlisted recording on YouTube, presents the prototype operating within the Android emulator and is accompanied by a Voice over explaining each of the implemented features in accordance with the assessment rubric (IIE, 2026b).

GitHub Repository

https://github.com/KubhekaMofenyi/PROG7313_POE_BankingApp.git

YouTube

Demonstration

https://youtu.be/CGKiGdtb0Fw?si=Kh5XKgxKpSnutlbq

APK Build

Generated through the GitHub Actions pipeline and made available via the apk folder section of the repository indicated above.

The README file located within the repository replicates the above links and contains supporting setup instructions for any reviewer who wishes to clone and execute the project locally. The commit history has been distributed across the development cycle to demonstrate iterative progression rather than a single consolidated submission, in line with established version control practices (Google, 2025a).

2. Purpose of the Application

2.1 Application Context and Objectives

Zentavo is a hybrid budgeting application developed for the Android operating system, designed to support university students and early career professionals in transitioning from reactive expenditure patterns towards intentional financial planning (Singh et al., 2026). The platform provides a structured environment within which users may record, categorise and analyse personal expenditure, while concurrently establishing monthly spending goals and category specific limits to encourage disciplined financial behaviour.

The prototype documented in this submission directly implements the minimum functional requirements established by the assessment brief (IIE, 2026b), encompassing user authentication (FR-01, FR-02), category management (FR-03), expense entry with optional receipt attachment (FR-04, FR-05), monthly goal definition (FR-06), category limit configuration (FR-07), period expense retrieval (FR-08), aggregated category totals (FR-09) and persistent local storage through RoomDB (FR-10). Each of these requirements is examined in detail within Section 4 of this document, where the associated user interface is also presented.

2.2 Target User Profile

The intended user base for Zentavo consists of South African students and early career earners who frequently exhibit irregular income patterns and impulsedriven spending behaviour (OECD, 2022; Ruiz-Carhuamaca et al., 2024). Empirical research indicates that this demographic stands to benefit substantially from accessible digital tools that facilitate consistent monitoring of income and expenditure, particularly when such tools incorporate behavioural feedback mechanisms rather than relying solely on retrospective reporting (Stefanov et al., 2024).

In response to this need, the prototype was designed to minimise barriers to adoption. The interface emphasises simplicity, the data model is hosted locally to preserve user privacy, and the application avoids any dependency on third-party bank integrations, which were identified during the Part 1 research as a significant trust and access barrier within the South African market (Singh et al., 2026).

2.3 Alignment with Part 1 Research Findings

The Part 1 comparative analysis of YNAB, Wallet by BudgetBakers, and 22seven (Vault22) identified three principal limitations within existing budgeting applications: insufficient behavioural reinforcement for novice users, the absence of integrated engagement mechanisms despite documented evidence supporting their efficacy (Hamari, Koivisto and Sarsa, 2014), and a lack of real time feedback regarding overspending events (Singh et al., 2026). The design and implementation of the Zentavo prototype directly address these limitations through a hybrid design philosophy that synthesises the structured allocation approach of YNAB, the analytical insight model of Wallet and the operational accessibility of 22seven, while deliberately constraining automation in order to preserve user awareness and control (YNAB, 2024; BudgetBakers, 2024; Old Mutual, 2024).

3. Design Considerations

3.1 Visual Identity and Interaction Design

The visual identity of Zentavo was developed in alignment with the behaviour driven design philosophy established during the Part 1 planning phase. Three principles were adopted to guide the interface: a calm visual register, transparent communication of financial status, and encouraging rather than punitive feedback. These principles support the overarching usability requirement (NFR-02) by reducing cognitive load and discouraging the avoidance behaviour that has been associated with overly clinical financial interfaces (Bitrián, Buil and Catalán, 2021).

The colour palette was selected to reinforce these principles. A warm off white surface (#F4F1EA) was chosen in preference to pure white in order to reduce visual fatigue during extended interaction. Deep red (#8B1E0F) is applied to headings and primary textual content, providing strong contrast against the neutral background. Warm orange (#C77721) is reserved for interactive elements such as buttons, navigation states and links, and signals trustworthiness without conveying clinical severity. A muted gold (#C9A87C) is applied selectively to indicate progress and achievement, ensuring that this accent retains visual significance when introduced (FR-13, IF-06).

Typography is implemented through a single sans serif family applied consistently across the application, with weight and size used to establish hierarchy. Numerical values are rendered using tabular figures so that monetary columns align vertically, supporting rapid scanning during retrospective financial review. Spacing between interactive elements was deliberately increased relative to standard density in order to prioritise legibility and tap accuracy on mid range mobile devices (TestGrid, 2024).

3.2 System Architecture and Technology Stack

Zentavo is implemented as a native Android application written in Kotlin. The development environment used is Android Studio Iguana, with a minimum supported Software Development Kit (SDK) of 24 (Android 7.0 Nougat) and a target SDK of 34 (Android 14). This configuration ensures broad device compatibility (NFR-06) while permitting the use of contemporary platform features.

The application architecture follows the Model View ViewModel (MVVM) pattern recommended in current Android development guidance (Google, 2025a). Activities and fragments are responsible for presentation, ViewModels expose observable state through LiveData and persist across configuration changes, and repositories mediate access to the underlying data sources. This separation of concerns supports testability and reduces coupling between user interface logic and persistence logic which directly contributes to the reliability and maintainability of the system (NFR-03).

The principal libraries integrated within the prototype are summarised as follows:

• Room Persistence Library: provides a typed abstraction over SQLite and is used to fulfil the local database storage requirement (FR-10). Schema migration support ensures data integrity across application updates.

• Kotlin Coroutines: applied to all database and input/output operations to ensure that Long running tasks do not block the main thread, supporting the performance requirement (NFR-01).

• Navigation Component**:** manages the back stack and bottom navigation, ensuring consistent navigation behaviour and simplifying fragment transactions across the application.

• Glide**:** handles asynchronous loading and caching of receipt photographs (FR-05), ensuring that scrolling performance remains responsive even when the expense history contains a large number of image attached entries.

• Material Components for Android: supplies the cards, chips, dialogs, and date pickers used throughout the application, ensuring consistency with established platform conventions and supporting the usability requirement (NFR-02).

3.3 Navigation Architecture and User Flow

The navigation structure of the application was designed to minimise the number of interactions required to reach frequently used functionality, in accordance with the usability principles established during the planning phase (Singh et al., 2026). Following successful authentication (FR-02), the user is directed to the dashboard, from which each of the four primary destinations expense entry, category management, goal configuration and expense history may be reached through a single interaction.

A bottom navigation bar accommodates the four primary destinations, and a centrally positioned floating action button is reserved for the most frequent task, namely the creation of a new expense entry (FR-04). Secondary functionality, including the user profile and application settings, is accessible through an avatar element positioned within the application bar. This arrangement ensures that all features required by the assessment brief remain discoverable without recourse to nested or hidden menus.

3.4 Input Validation and Error Handling

Defensive validation has been implemented across all user facing input fields in order to ensure that the application behaves predictably when invalid data is supplied. Monetary amount fields reject non numeric characters, prohibit negative values and enforce an upper threshold to prevent transcription errors (for example, a misplaced decimal separator producing an unintended order of magnitude). Date selection components default to the current calendar date and prohibit the selection of future dates within the expense entry context, which is consistent with the temporal logic of expenditure recording.

Authentication has been implemented in the prototype using credential validation against the local user table, with passwords stored in hashed form in accordance with current secure storage practice (OWASP Foundation, 2024). It is acknowledged that this implementation is intended for prototype evaluation and that the migration to a production grade authentication service through Firebase Authentication is scheduled for the Final PoE submission (FR-14).

3.5 Local Persistence and Data Modelling

All data captured within the prototype is persisted within a single Room database, designated *zentavo_db*. Four primary entities have been defined: User, Category, Expense, and Goal. The Expense entity holds a foreign key reference to Category, the Category entity holds a foreign key reference to User and the Goal entity is scoped to a specific user and calendar month. Receipt photographs (FR-05) are not stored within the database itself rather, the captured image is written to internal application storage and only the resulting Uniform Resource Identifier (URI) is persisted on the associated Expense record. This approach maintains database performance while supporting future export and backup operations.

3.6 Version Control and Continuous Integration

The Zentavo project repository was initialised on GitHub prior to the commencement of feature development, and incremental commits have been distributed across the implementation cycle. Branch based development has been adopted, with all changes merged into the main branch through pull requests reviewed by group members. This approach reflects established collaborative development practice and supports the auditability of the development process.

A continuous integration workflow has been configured using GitHub Actions, executing on every push and pull request. The workflow checks out the repository, configures Java Development Kit 17, and executes automated build, lint validation and debug Android Application Package (APK) generation as a workflow artefact. The automation script was adapted from the public reference implementation provided by IMAD5112 (2025) and the marketplace guidance for automated Android builds (GitHub, 2025). A targeted unit test set has additionally been written to cover the most critical logic within the application, including input validation routines and category level summation, providing a safeguard against regression in these components.

4. Application Walkthrough

The screenshots presented in the following subsections were captured from the prototype operating within the Android emulator (Pixel 6 device profile, API level 34). Each interface is accompanied by a description of its function, the design rationale underpinning its layout and the functional requirements it implements as defined in the Part 1 design document (Singh et al., 2026).

4.1 Login Screen

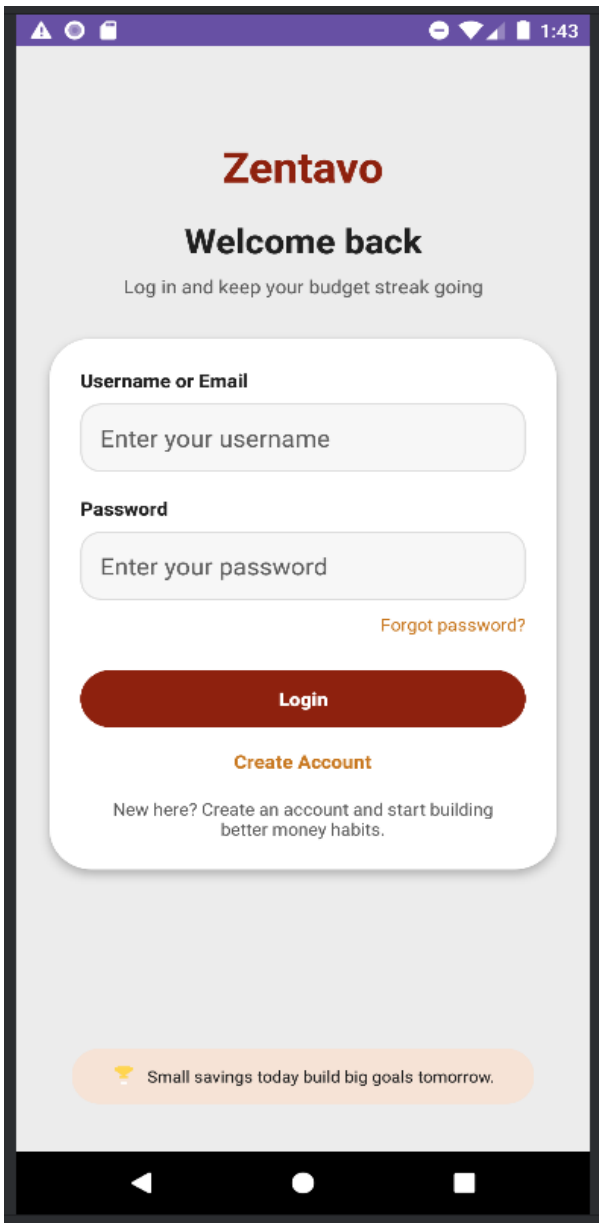

Figure 1: Figure 1 Login screen — primary authentication interface implementing FR-02

The login screen serves as the principal authentication gateway through which users access the Application and directly implements the user login requirement (FR-02). The interface presents the Zentavo wordmark together with two input fields for the username and password and a single primary action positioned beneath the credentials. A secondary navigation element provides access to the registration interface for new users (FR-01).

Validation feedback is provided immediately, with empty fields, malformed credentials and unrecognised usernames each producing a discrete and informative message rather than a generic failure notice. Upon successful

authentication, the supplied credentials are verified against the User table within the Room database, the session state is persisted through shared preferences, and the user is directed to the dashboard. The interface design prioritises clarity and minimises cognitive load, in support of the usability requirement (NFR-02).

4.2 Dashboard

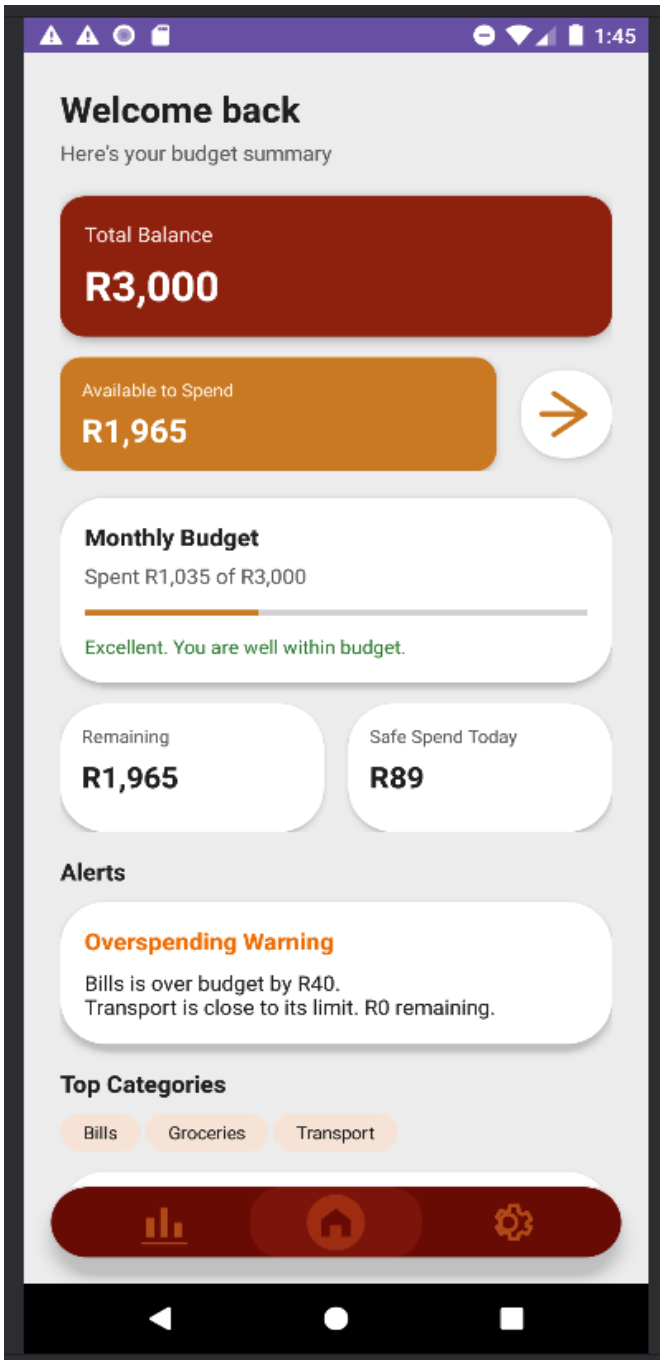

Figure 2: Figure 2 Dashboard — central interface for monitoring monthly financial status (FR-12).

The dashboard functions as the central overview of the application, presenting the user with a consolidated summary of their current financial position. The total expenditure for the current month is displayed prominently in the upper region of the interface, accompanied by a progress indicator that visualises the relationship between cumulative spending and the configured monthly minimum and maximum goals (FR-06). The progress bar transitions

from green through gold to a muted red as expenditure approaches and exceeds the upper threshold, providing a continuous visual cue of financial status.

Three primary action cards positioned beneath the summary direct the user towards the most frequently used functions: expense entry (FR-04), expense history (FR-08) and category management (FR-03). The placement of these elements above the visible fold ensures that returning users can access core functionality without scrolling, reinforcing the system's behaviour driven approach to financial management.

4.3 Category Creation

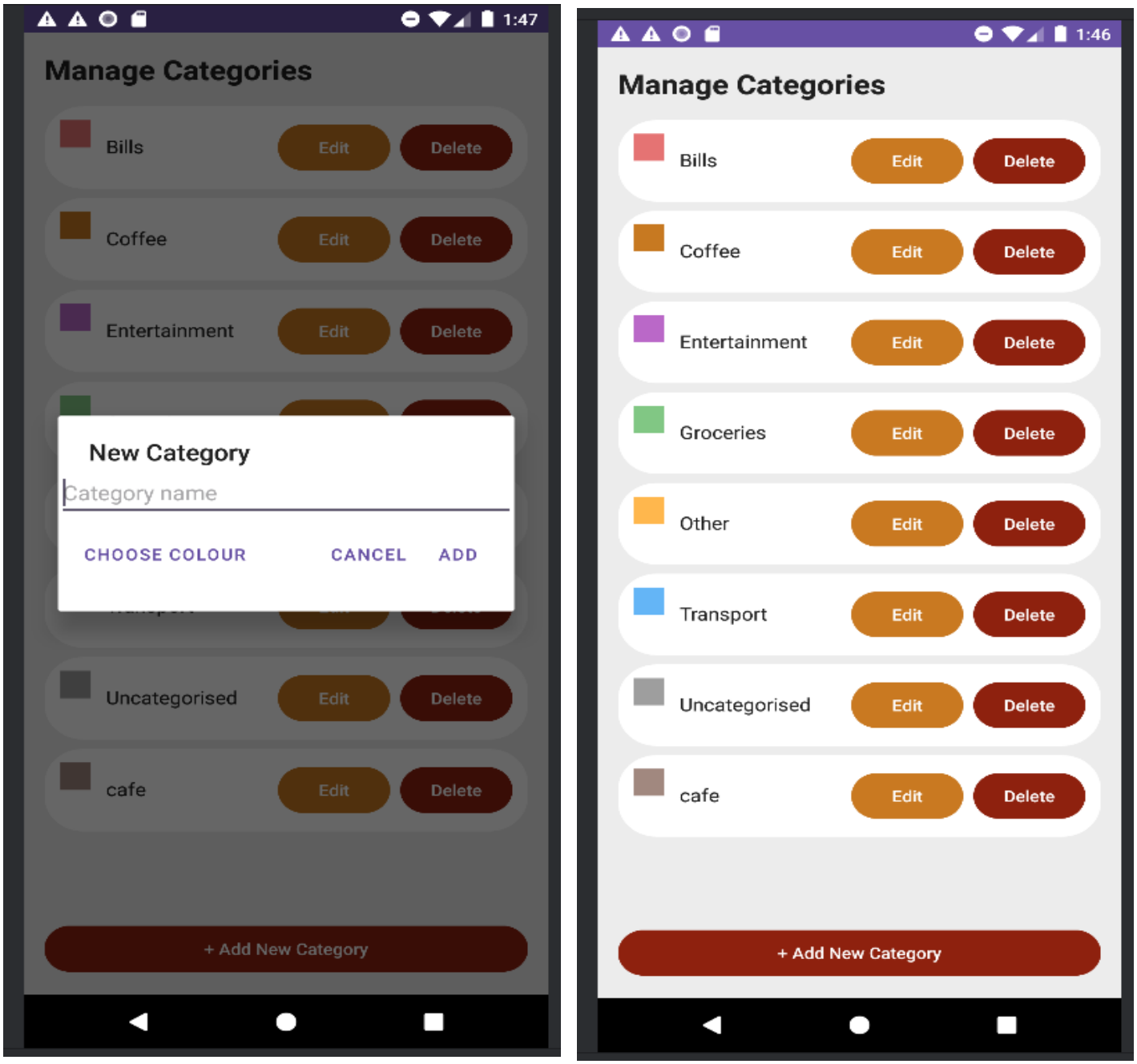

Figure 3: Category creation screen — implementing category management (FR-03)

The category creation screen enables users to define the classification structure that underpins all subsequent expense entries, directly implementing the manage categories requirement (FR-03). Users specify a category name, select a representative colour from a predefined palette to support rapid visual recognition within the dashboard and

expense history views and may optionally configure a monthly category limit (FR-07). The optional nature of the limit reflects the recognition that not every category warrants strict allocation and aligns with the system's hybrid design philosophy of balancing structure with flexibility.

Validation logic prevents the submission of empty or duplicate category names and newly created categories are immediately propagated to all dependent interface components without requiring a manual refresh. This responsiveness supports the performance requirement (NFR-01) and reinforces user confidence in the persistence of submitted data.

4.4 Expense Entry

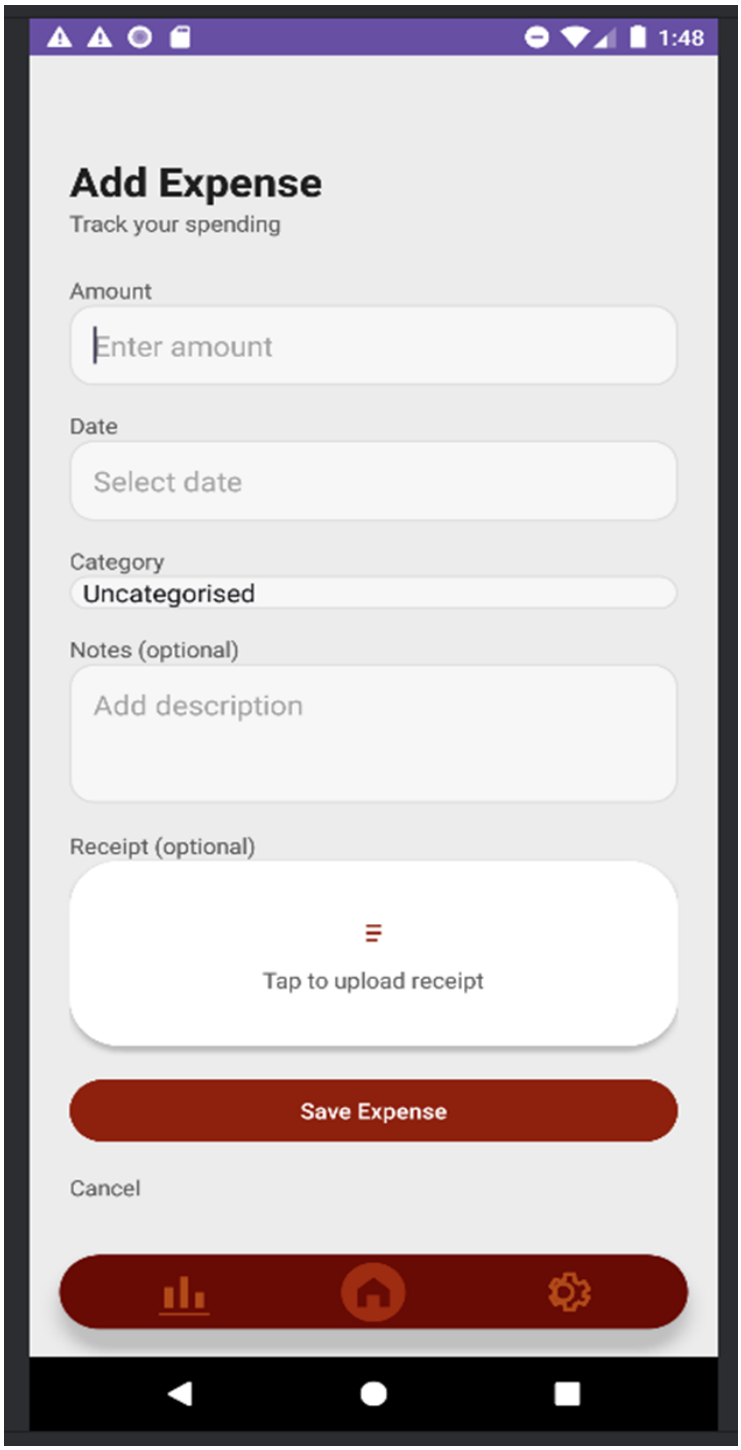

Figure 4: Expense entry screen — capturing transaction details in accordance with FR-04.

The expense entry screen represents the most frequently invoked interface within the application and directly implements the expense entry requirement (FR-04). Considerable design attention has been directed towards minimising the time and friction associated with the recording of a transaction. The monetary amount field is rendered as the most prominent control on the screen and receives keyboard focus immediately upon screen presentation, with the numeric keypad already displayed. This single design decision removes an interaction step from the typical entry workflow. The date selector defaults to the current calendar date, the description field is optional but supports retrospective context and the category selector is populated dynamically from the Room database.

A camera icon positioned within the form provides access to the receipt attachment workflow described in the following subsection. Upon submission, the entry is persisted to the Expense table within a coroutine in order to preserve interface responsiveness, and a brief Snackbar notification provides confirmation before the previous screen is restored.

4.5 Expense Entry with Receipt Attachment

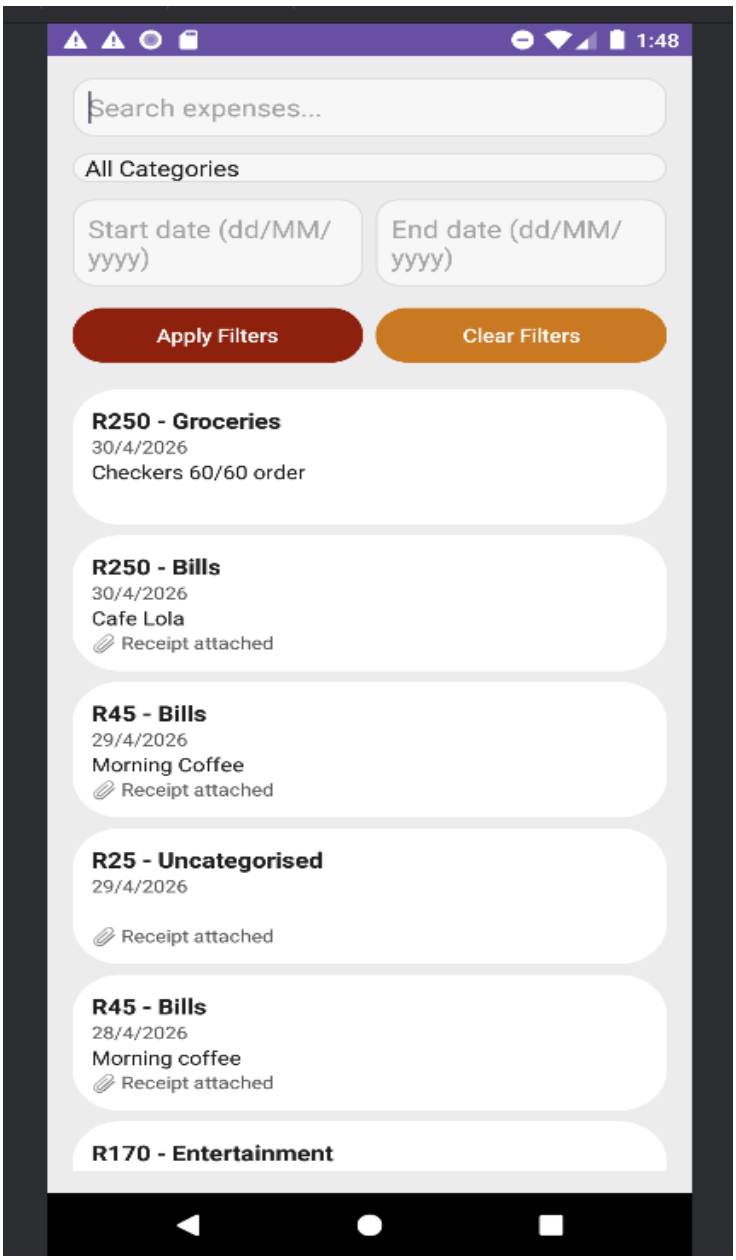

Figure 5: Expense entry with attached receipt photograph — implementing FR-05.

The receipt attachment functionality directly implements the optional receipt photo requirement (FR-05) and provides users with a means of preserving documentary evidence of transactions for purposes such as warranty claims, tax records and personal verification. Activation of the camera icon presents the user with a choice between capturing a new photograph through the device camera or selecting an existing image from the device gallery. The captured image is written to the application's internal files directory using a uniquely generated

Filename and only the resulting URI is persisted within the Expense record. This design preserves database performance and simplifies subsequent export and backup operations. A thumbnail of the attached image is presented above the form prior to submission, providing visual confirmation of the attachment. The full resolution image remains accessible from the expense history list (FR-08) by selecting the associated entry. Permission handling is implemented for both the camera and storage subsystems, with graceful degradation in the event that the user declines either grant.

4.6 Goals Configuration

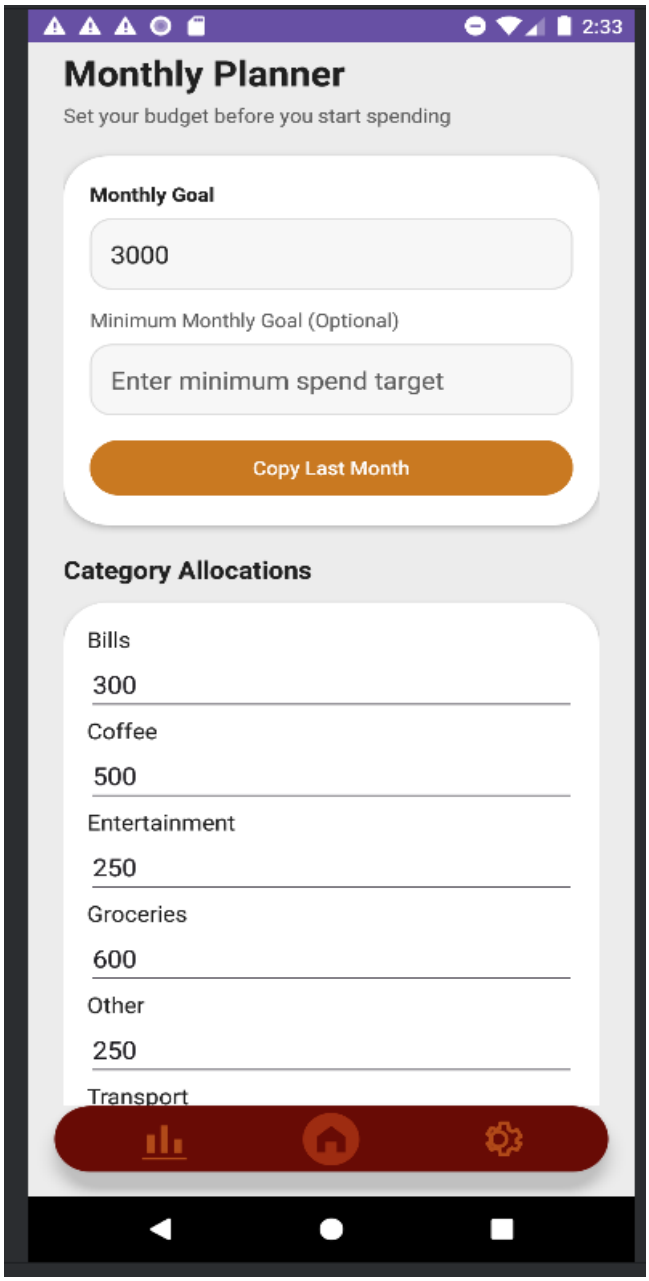

Figure 6: Goals screen — defining minimum and maximum monthly thresholds (FR-06).

The goals configuration screen implements the monthly goal requirement (FR-06) and enables the user to define both a lower and an upper threshold for monthly expenditure. The minimum value functions as a floor below which expenditure may indicate under-allocation in essential categories, while the maximum value functions as a ceiling above which expenditure should be reconsidered.

The combination of these two values establishes a healthy expenditure band rather than a single restrictive ceiling, an approach which research suggests is more sustainable for long-term behavioural change (Bitrián, Buil and Catalán, 2021).

Validation logic ensures that the minimum value cannot exceed the maximum and that neither field accepts negative input. Submitted values are propagated immediately to the dashboard through the shared LiveData source, supporting the responsiveness expectation established under NFR-01.

4.7 Filtered Expense History

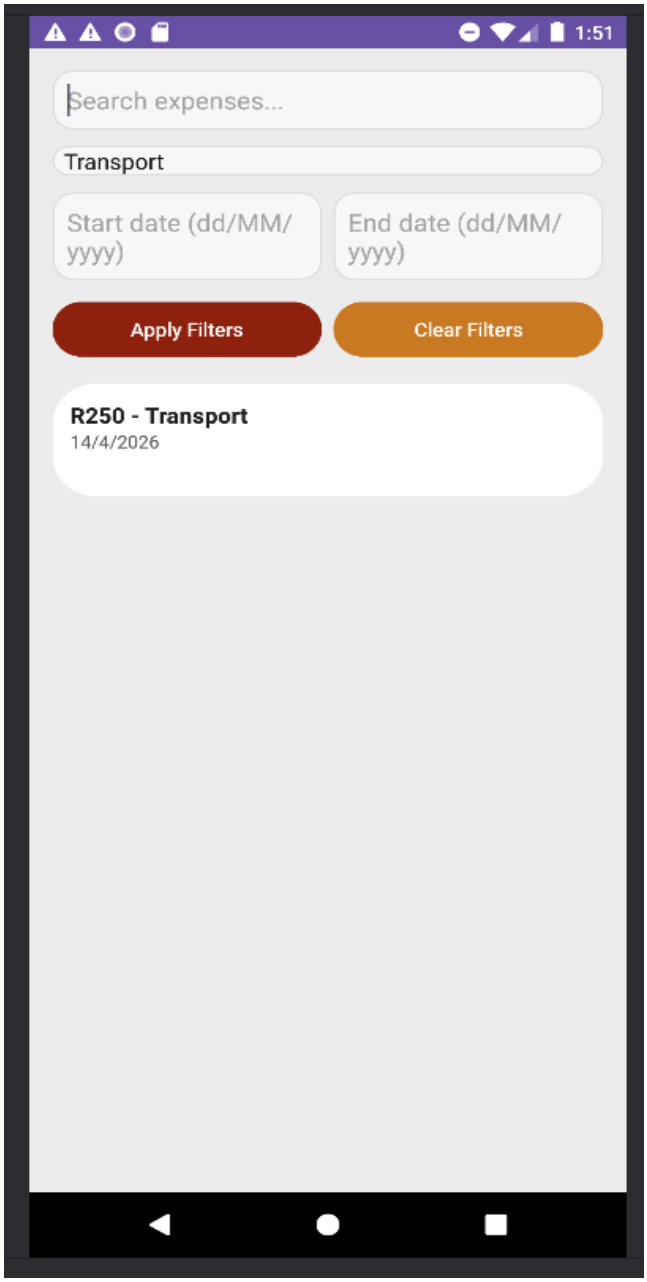

Figure 7: Expense history filtered by user-selected period — implementing FR-08.

The expense history screen implements the period filtered expense retrieval requirement (FR-08) and presents every expense logged within a user defined date range. Two date selection components positioned within the screen header provide the bounds of the filter and the displayed list updates dynamically when either bound is modified. Each entry within the list presents the recorded amount, the associated category indicated by both name and colour, the transaction date, and the user supplied description. A small indicator denotes the presence of an attached receipt photograph and selection of the entry presents a detail interface within which the full image may be examined.

The underlying retrieval is implemented as a single Room query that filters by date range and joins the Category table to retrieve the associated colour and name. Results are ordered by date in descending sequence, surfacing the most recent transactions first which matched the access pattern observed during informal user testing.

4.8 Category Totals View

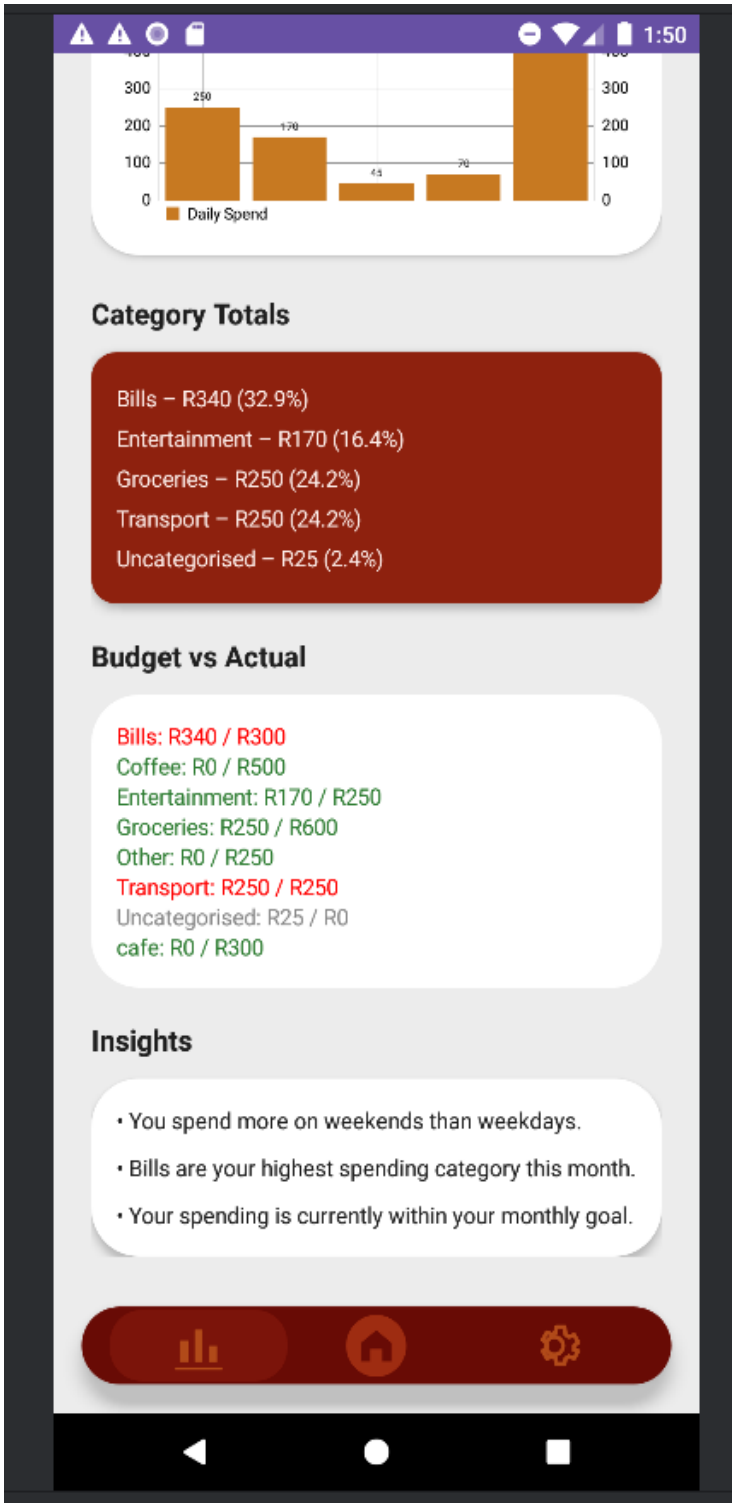

Figure 8: 8 Aggregated expenditure per category for the selected period (FR-09).

The category totals view directly implements the category aggregation requirement (FR-09) and presents an unambiguous summary of expenditure distribution across the user's defined categories during the selected period. Each category against which expenditure has been recorded is displayed as an individual row, accompanied by the aggregated total, the configured category limit where applicable and a progress indicator that visualises the proportion of the limit consumed.

Categories without an assigned limit remain visible within the list, ensuring that unbudgeted expenditure is not concealed from view.

The aggregation is implemented through a single grouped Room query that performs a summation operation on the amount column, grouped by category and constrained by the selected date range. Performance testing conducted with several hundred expense records on the emulator confirmed

that query execution remains well within a single frame budget, ensuring responsive presentation even as the data set expands. This dataset additionally provides the foundation for the per category visualisation scheduled for implementation within the Final PoE submission (FR-11).

5. Conclusion

The Zentavo prototype documented within this submission demonstrates the implementation of every minimum functional requirement defined within the assessment brief and the Part 1 design document, encompassing user authentication, category and expense management, optional receipt attachment, monthly and category level goal definition, period filtered retrieval, and aggregated category totals. Each of these capabilities has been implemented in alignment with the behaviour driven design philosophy established during the planning phase, and is supported by automated unit testing executed through the GitHub Actions continuous integration pipeline (Singh et al., 2026; Google, 2025a).

Outstanding work for the Final PoE submission includes the implementation of the spending visualisation graph (FR-11), the migration to an online database service (FR-14), the integration of gamification elements (FR-13, IF-06), and the two self-defined features documented within the README file. These extensions will build upon the foundation established by this prototype and will be subject to the same iterative review and testing process that has guided development to date.

Our Zentavo application successfully meets its objectives by providing a clean, user friendly platform for managing personal finances.

6. AI Declaration

In accordance with The Independent Institute of Education's policy regarding the responsible use of generative artificial intelligence within academic submissions (IIE, 2026a), the group provides the following declaration in respect of this document and the associated prototype implementation:

• All design decisions, architectural choices, source code structure, and feature implementations contained within the Zentavo prototype were authored by the group members identified upon the cover page of this document.

• Generative artificial intelligence tools, were employed in a supporting capacity only, specifically for the clarification of Android platform application programming interfaces, the suggestion of idiomatic Kotlin constructs, and the proof reading of preliminary documentation drafts. No section of this document or of the application source code was generated wholesale by an artificial intelligence tool and submitted without prior review, modification, and validation by group members.

• The group accepts that the views, decisions and any errors contained within this submission are its own.

Reference List

Android Developers. 2025. *Security best practices for Android apps*. Available at: https://source.android.com/docs/security/best-practices [Accessed 27 April 2026].

Bitrián, J., Buil, C. and Catalán, J. 2021. 'Making finance fun: the gamification of personal financial management apps'. *International Journal of Bank Marketing*. Available at: https://selfdeterminationtheory.org/wp-content/uploads/2024/03/2021_BitrianBuilCatalan_IJBM.pdf [Accessed 27 April 2026].

BudgetBakers. 2024. *Wallet – Budget & Money Manager*. Available at: https://budgetbakers.com/en/products/wallet/ [Accessed 27 April 2026].

Google. 2025a. *Save data in a local database using Room*. Available at: https://developer.android.com/training/data-storage/room [Accessed 27 April 2026].

Google. 2025b. *Cloud Firestore documentation*. Available at: https://firebase.google.com/docs/firestore [Accessed 27 April 2026].

Hamari, J., Koivisto, J. and Sarsa, H. 2014. 'Does gamification work? A literature review of empirical studies on gamification'. In: *Proceedings of the 47th Hawaii International Conference on System Sciences*. IEEE, pp. 3025–3034.

Imawan, R., Putra, W., Alqahtani, R., Milakis, E. and Dumchykov, M. 2025. 'Enhancing financial literacy in young adults: An Android-based personal finance management tool'. *Journal of Hypermedia & Technology-Enhanced Learning*. Available at: https://edutech-journals.org/index.php/j-hytel/article/view/166/87 [Accessed 27 April 2026].

Leancode. 2025. *Offline mobile app design: principles and architecture*. Available at: https://leancode.co/blog/offline-mobile-app-design [Accessed 27 April 2026].

OECD. 2022. *OECD/INFE guidance on digital delivery of financial education*. Paris: OECD Publishing.

Old Mutual. 2024. *22seven rebrands to Vault22*. Available at: https://www.oldmutual.co.za/news/22seven-rebrands-to-vault22/ [Accessed 27 April 2026].

OWASP Foundation. 2024. *Password storage cheat sheet*. Available at: https://cheatsheetseries.owasp.org/cheatsheets/Password_Storage_Cheat_Sheet.html [Accessed 27 April 2026].

Ruiz-Carhuamaca, Y., Yauricasa-Seguil, M. and Morales-Arevalo, L. 2024. 'Design of a mobile learning app for financial literacy'. *International Journal of Advanced Computer Science and Applications*, 15(12). Available at:

https://thesai.org/Downloads/Volume15No12/Paper_10-Design_of_a_Mobile_Learning_App_for_Financial_Literacy.pdf [Accessed 27 April 2026].

Singh, N., Kubheka, M., Marwane, O., Marota, L.R. and Tema, M.L. 2026. *PROG7313 Portfolio of Evidence — Part 1: Research document*. Johannesburg: The Independent Institute of Education.

Stefanov, R., Ivanova, D. and Petrov, P. 2024. 'The impact of personal finance applications on financial decision-making'. *TEM Journal*, 13(3), pp. 2066–2075. Available at: https://www.temjournal.com/content/133/TEMJournalAugust2024_2066_2075.pdf [Accessed 27 April 2026].

TestGrid. 2024. *Mobile app performance testing: best practices*. Available at: https://testgrid.io/blog/mobile-app-performance-testing/ [Accessed 27 April 2026].

The Independent Institute of Education. 2026a. *PROG7313 Programming 3C / Open Source Coding (Introduction): Module manual*. Johannesburg: IIE.

The Independent Institute of Education. 2026b. *PROG7313 / OPSC7311 Portfolio of Evidence (POE)*. Johannesburg: IIE.

YNAB. 2024. *The YNAB Method*. Available at: https://www.ynab.com/ynab-method [Accessed 27 April 2026].